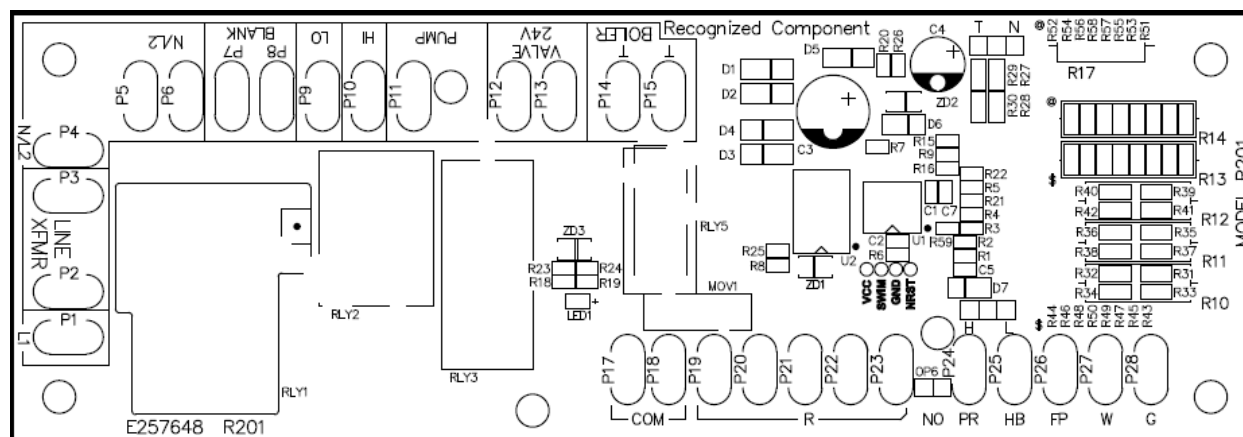


APPLICATION

The R201 multi-function control board provides microprocessor control of a “combo” air handler. It provides 2-speed line voltage control of a fan motor, line voltage control of a water pump, 24Vac control for an isolation valve and control of a dry switching relay to turn on/off a water heater. A flashing green LED indicates that the microprocessor is operating correctly. An auto pump timer cycles the pump daily for periods of extended absences or lack of pump use.

Figure 1. R201 Board.



Note: The R201 board must be factory installed and wired inside the equipment it is controlling.

FEATURE

Heating Cycle (Blower on High or Low, depending on jumper; High if HB input is present)

1. Automatic pump timer: The circulating pump automatically energizes, to purge the hot water coil and piping four times daily for 60 seconds each cycle. The pump timer cycle is suppressed while the G signal is ON and will resume after the G signal is removed.
2. Blower-on delay: On a call for heat, the hot water coil is preheated for 90 seconds before the blower energizes.
3. Blower-off delay: Blower continues to operate for 20 seconds after thermostat is satisfied, extracting more heat from the heating coil and increasing heating performance.

Cooling Cycle (Blower on High)

1. Blower-off delay: Blower continues to operate for 45 seconds after thermostat is satisfied, extracting more cooling from the cooling coil and maximizing cooling efficiency.

Optional Features (requires an optional field installed external switch)

1. Freeze protection: The circuit board has terminals for an optional freeze protector switch. Freeze protection energizes the circulating pump if the water coil temperature falls below 40 degrees. This feature continues to operate for 30 seconds after the thermostat is satisfied.
2. Cold weather fan control: The circuit board has terminals for an optional cold weather fan switch (Heater Booster input). When the outdoor temperature falls below the fan control set point, the fan control switches the blower from low to high speed.
3. Priority Control: Circuit board has terminals for optional priority switch. The priority switch de-energizes during periods of hot water domestic draw.

GENERAL SPECIFICATION

Line Voltage Terminals

Table 1. Line Voltage Terminals.

Terminal	Description	Purpose / Rating
P1	L1	Line Hot, 120 VAC, 50/60 Hz
P2, P3	LINE XFMR	Transformer Primary Leads, 120 VAC, 50/60 Hz
P4, P5, P6	N/L2	Line Neutral, 120 VAC, 50/60 Hz
P7, P8	BLANK	Not in use
P9	LO	Blower Low Fan Speed Output, 120 VAC, 50/60 Hz, 16A Resistive Load
P10	HI	Blower High Fan Speed Output, 120 VAC, 50/60 Hz, 16A Resistive Load
P11	PUMP	Circulating Pump Output, 120 VAC, 50/60 Hz, 5A Resistive Load

Low Voltage Terminals

Table 2. Low Voltage Terminals.

Terminal	Description	Purpose / Rating
P12, P13	24V VALVE	Motorized Valve or Pump Relay Output, 24 VAC, 5A Resistive Load (P12- 24VAC Hot; P13- Common)
P14, P15	T T	Dry Contact Boiler On/Off Output, 24 VAC, 5A Resistive Load
P16	COND	Not in use
P17, P18	COM	Common, 24 VAC
P19, P20	R	Hot, 24 VAC
P21, P22	R	Hot, 24 VAC
P23	R	Hot, 24 VAC
P24	PR	Priority Input
P25	HB	Heater Booster Input, 24 VAC
P26	FR	Freeze Protection Input, 24 VAC
P27	W	Thermostat Heat Input, 24 VAC
P28	G	Thermostat Fan Input, 24 VAC

LED Indication

Flashing Green- Microprocessor working correctly.

Steady Green – Microprocessor failed.

Off – Low voltage - below 16 VAC (+/- 3 VAC)

Other Specifications

Operating Temperature - -20 to 140 °F (-28 to 60 °C)

Storage Temperature - -30 to 150 °F (-34 to 65 °C)

Operating Humidity – 5 to 90% RH, non-condensing

Operating Life – 100K cycles at full load

Agencies – Tested to UL 873

Board Jumper Options

- OP6 – Circuit board tab to activate Priority switch input.
- Blower Speed Option for Heating – L for Low fan speed (default); H for High fan speed (overridden by HB input, if present, to High)
- Test Mode Option Jumper – N to middle pin (default from factory) - or no jumper: Normal operation mode; T- to middle pin for test mode.

Dimensions

(W x D x H): 6.85 x 2.40 x 1.65 inch

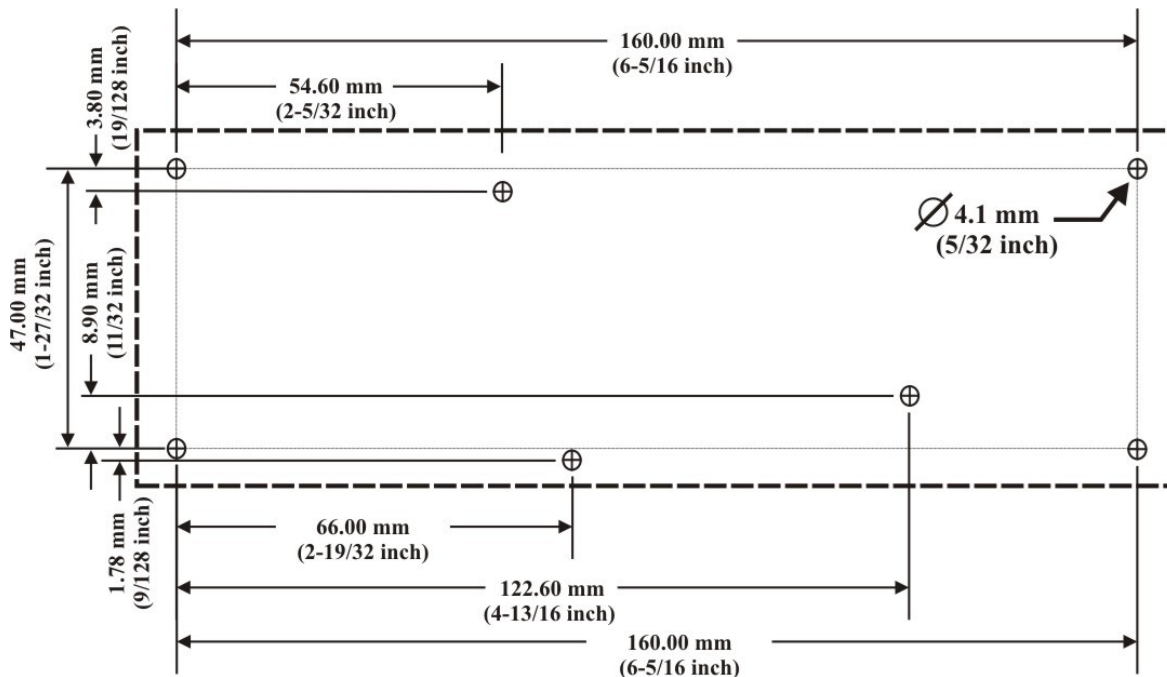
INSTALLATION



This device must be installed inside an electrical enclosure with minimum clearances to grounded metal in accordance with UL 1995 to reduce the risk of unintentional contact with an uninsulated live part. It must be protected by a maximum of 20A fuse or circuit breaker.

Mark and drill 7 mounting holes with the same diameter and distance as described below. Align board and the plastic spacers over the holes, firmly push the board until all spacers snap into the mounting holes.

Figure 2. Mounting Dimension.



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